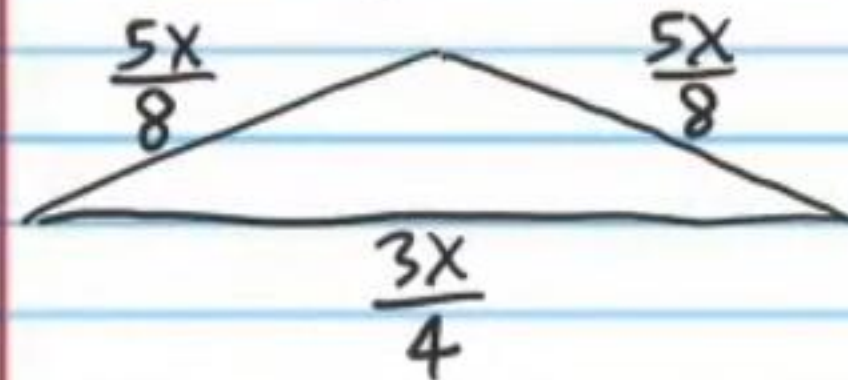


Applications of Algebra (Part I) Homework

Name
Date
Course

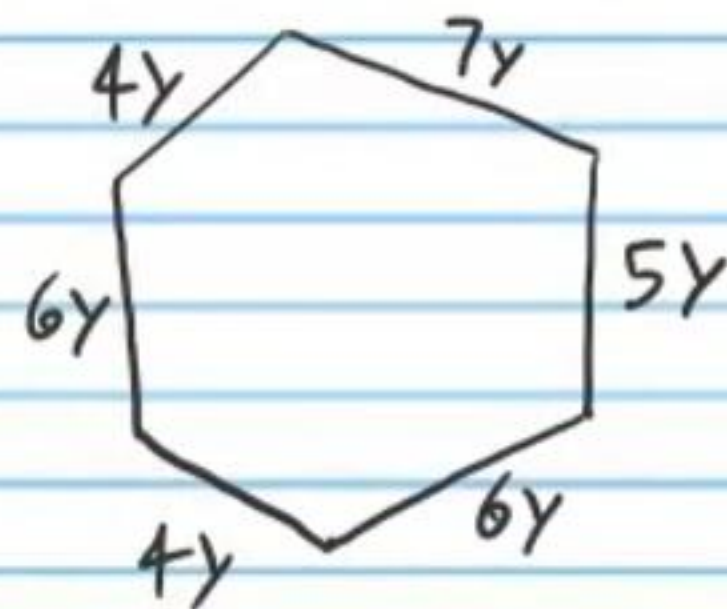
Find an expression for each of the values below.

- ① The perimeter of the triangle below.



$2x$

- ② The perimeter of the hexagon below.



$32y$

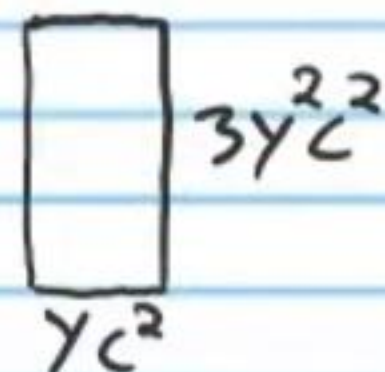
- ③ The perimeter of a rectangle with length $2.3z$ and width $0.99z$.

$6.58z$

- ④ The perimeter of a square with side length $2x$. Remember, all sides of a square have equal length.

$8x$

- ⑤ The area of the rectangle below.



$3y^3c^4$

- ⑥ The area of a triangle with height $3.6x$ and base $2.008x$.

$3.6144x^2$

- ⑦ The area of a square with side length $\frac{6xy}{7z^3}$.

$\frac{36x^2y^2}{49z^6}$

- ⑧ The area of a triangle with base $\frac{3x}{5}$ and height $\frac{3x}{4}$. Reference problem #12 in class.

$\frac{3x^2}{20}$

Find an expression for the values below (keep π as a factor).

- ⑨ The volume of a rectangular prism with length $6x^3$, width $2x$, and height $3x^2$.

$36x^6$

- ⑩ The area of a circle with radius $7y^3$.

$49\pi y^6$

- ⑪ The surface area of a rectangular prism with length $9z$, width z , and height $3z$.

$78z^2$

- ⑫ The area of a parallelogram with base $5.23x^2yz^2$ and height $2.03x^4y^2z^4$.

$10.6169x^6y^3z^6$

- ⑬ The area of a trapezoid with first base $3y$, second base y , and height $2x^5y$.

$4x^5y^2$

- ⑭ The volume of a cylinder with diameter $\frac{7x}{2}$ and height $4x^2$.

$\frac{49\pi x^4}{4}$

- ⑮ The volume of a sphere with radius $3z^7$.

$36\pi z^{21}$

⑩ The surface area of a sphere with radius $8xyz$.

$$256\pi x^2 y^2 z^2$$

⑪ The circumference of a circle with diameter $14x^5yz$.

$$14\pi x^5 yz$$

Evaluate. Show the formula that you use and be sure to use the new versions, without "x" or "÷" signs.

⑫ Find the area of a parallelogram with base 2 feet, and height 8 feet.

$$16 \text{ ft}^2$$

⑬ Find the volume of a cylinder with radius 5 inches and height 3 inches (keep π as a factor).

$$75\pi \text{ in}^3$$

⑭ Find the area of a trapezoid with bases 6 cm and 1 cm, and height 3 cm.

$$10.5 \text{ cm}^2$$

⑮ Find the volume of a rectangular prism with length $\frac{5}{6}$ miles, width $\frac{2}{3}$ miles, and height $\frac{1}{2}$ miles.

$$\frac{5}{18} \text{ mi}^3$$

⑯ Find the surface area of the rectangular prism in the previous problem.

$$\frac{47}{18} \text{ mi}^2$$

⑰ Evaluate your answer from problem # 5 if $c=1$ and $y=2$.

$$24$$

⑱ Evaluate your answer from problem # 8 if $x=3$.

$$\frac{27}{20}$$

⑲ Evaluate your answer from problem # 12 if $x=1$, $y=3$, and $z=1$. You can use a calculator.

$$286.6563$$

⑳ Find the amount of money in Linsey's new bank account if she deposits $P=489$ dollars and earns simple interest at $r=0.07$ rate paid yearly for $t=6$ years. Use the following formula: $A = P + Prt$.

$$\$694.38$$

㉑ Tim's salary decreased by $r=0.13$. His original salary was $F = \$29,144$. What is his new salary? Use the following formula: $N = F - Fr$.

$$\$25,355.28$$